Anchor-Guided Dequantization for Federated Learning

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Abstract

Federated learning systems often mix clients with very different hardware budgets. Some clients can transmit full-precision model updates, while others must aggressively quantize before upload. That mismatch introduces systematic bias into server aggregation. We study a server-side correction approach, AGD-Smart, that uses full-precision *anchor* clients to supervise a residual CorrectionNet for low-bit client deltas. The implementation combines client-to-anchor smart matching, a replay buffer over recent rounds, optional distillation on a server validation split, warmup before correction is applied, and direction-preserving blending at inference time. Across the reported configurations, AGD-Smart consistently outperforms the simplified ProWD baseline by 1.9–5.7 percentage points. Its strongest gains appear under severe INT2 quantization, where it outperforms FedAvg by +14.2 points at round 10 and remains ahead by +2.1 points at round 50.

1 Introduction

Federated learning (FL) trains a shared model across many clients without centralizing raw data [McMahan et al., 2017]. In practice, however, participating devices are rarely homogeneous. A laptop-class client may upload a full-precision update, while an embedded device or phone may need INT8, INT4, or even INT2 quantization to fit its communication and memory budget. Once those updates are averaged together, quantization error no longer behaves like benign noise; it becomes a structured bias in the aggregated direction.

This project studies a simple question: can a server learn to reverse some of that bias if a small subset of clients still provide clean, full-precision updates? We use those full-precision clients as *anchors*. Their updates provide round-level supervision for a server-side CorrectionNet that maps a quantized delta toward a more full-precision-like delta before aggregation.

That question sits at the intersection of two existing FL lines of work. Heterogeneity-aware optimizers such as FedProx, SCAFFOLD, and FedNova study how client drift and uneven local progress affect aggregation [Li et al., 2020, Karimireddy et al., 2020, Wang et al., 2020]. Mixed-precision and bit-width-aware methods such as ProWD, FedShift, and recent client-specific quantization schemes instead focus on what happens when clients cannot all communicate at the same precision [Yoon et al., 2022, Seo et al., 2024, Chen and Vikalo, 2024].

The current implementation goes beyond a naive dequantizer in several ways. Instead of training on only the current round, it keeps a replay buffer over the most recent rounds. Instead of forcing every low-bit client toward the same anchor mean, it can assign each constrained client its most similar anchor by cosine similarity. It also supports an optional distillation term computed on a server validation set and blends corrected deltas with the original quantized delta to preserve client-specific direction under non-IID data.

Our main contributions are:

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

- A template-based writeup of the implemented anchor-guided correction pipeline, including warmup, replay buffering, smart matching, normalization, and optional similarity-aware aggregation.
- A reusable architecture diagram that makes the per-round data flow and supervision path explicit.
- An empirical summary showing that the method is most useful in the severe quantization regime, especially INT2, while still improving over the simplified ProWD baseline in every reported configuration and matching or beating FedShift on the shared settings, with the clearest gains at early INT2, final INT4, and non-IID INT8.

2 Related Work

FedAvg remains the standard FL baseline because of its simplicity and strong performance when client updates are already directionally compatible [McMahan et al., 2017]. Later optimization work made the heterogeneity problem more explicit. FedProx adds proximal regularization to stabilize training when local objectives and available local work differ across clients [Li et al., 2020]. SCAFFOLD corrects client drift with control variates [Karimireddy et al., 2020], while FedNova analyzes objective inconsistency under unequal local progress and fixes it through normalized aggregation [Wang et al., 2020]. These methods are important context because they show that even before quantization, federated aggregation is already sensitive to client mismatch.

Communication-efficient learning introduces another axis: compression. QSGD is an influential quantized optimization method in distributed training [Alistarh et al., 2017], and FedPAQ brings periodic averaging plus quantized message passing into the FL setting [Reisizadeh et al., 2020]. More recent work pushes this further toward resource-aware mixed precision. For example, Chen and Vikalo propose client-specific mixed-precision quantization for resource-constrained heterogeneous devices [Chen and Vikalo, 2024]. Those papers primarily study how to train and communicate efficiently under low precision, whereas this project focuses on repairing the low-bit uploads after they arrive at the server.

The closest prior work is ProWD, which explicitly targets bit-width heterogeneous FL through a server-side dequantizer [Yoon et al., 2022]. FedShift addresses the same mixed-precision aggregation problem from a different angle by applying a per-layer global mean shift that aligns quantized uploads to the anchor distribution before aggregation [Seo et al., 2024]. That makes FedShift an especially useful comparison: it adds a real correction step without any learned model, replay buffer, or client-specific target selection.

Our implementation sits in this same mixed-precision corridor, but it differs in three practical ways: it uses full-precision anchor clients as direct supervision, it trains the corrector on a replayed history instead of only the current round, and it uses smart client-to-anchor matching rather than a single global target. In the shared experiments, that extra structure matters. FedShift is already a stronger lightweight baseline than ProWD in most near-IID settings, yet AGD-Smart still pulls ahead where personalized correction matters most: early INT2 convergence, final INT4 accuracy, and robustness under non-IID data. The result should therefore be read as a server-side, anchor-supervised dequantization variant positioned between plain FedAvg and recent mixed-precision correction methods.

3 Method

3.1 Setting

At each communication round, the server broadcasts the current global model to all clients. Anchor clients return full-precision updates, while constrained clients return low-bit quantized updates. The server's job is to repair the constrained uploads before aggregation without disturbing the clean anchor signal.

3.2 Target Construction

The correction module operates only on *learnable* parameters. BatchNorm running statistics bypass the correction path unchanged. Each client delta is flattened into a single vector before training the corrector.

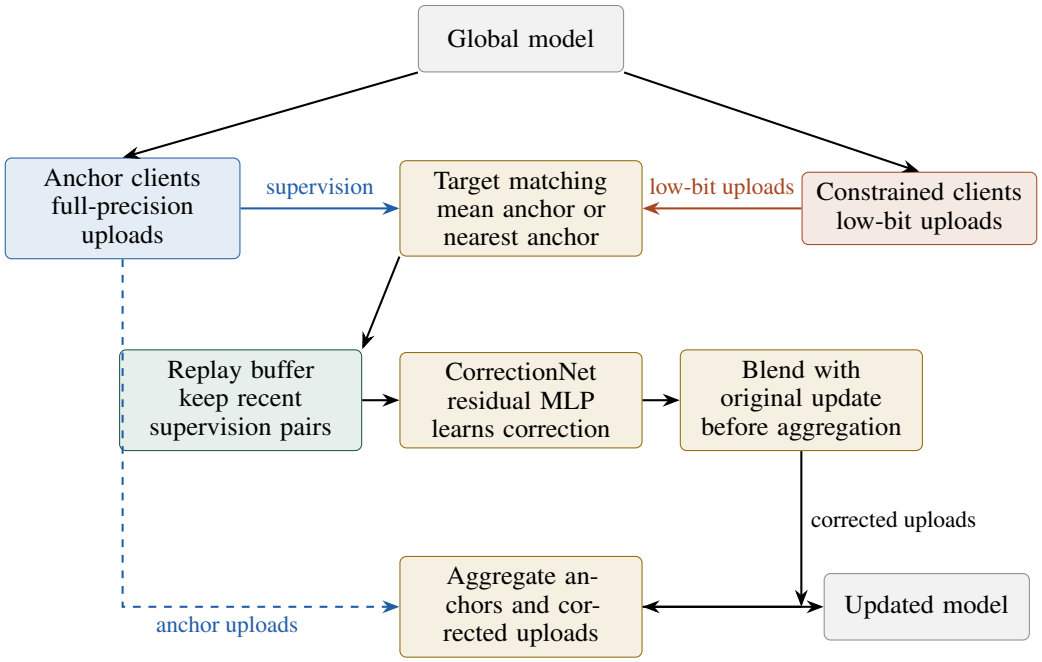


Figure 1: Implemented round-level pipeline for anchor-guided dequantization. Full-precision anchor clients supervise a server-side correction model. The server matches constrained uploads to anchor targets, reuses recent supervision through a replay buffer, applies a learned correction, blends it with the original low-bit update, and aggregates the repaired uploads with the anchors.

Two target modes are implemented. In the simpler version, every constrained client is trained toward the mean anchor update for that round. In the smart matching version, each constrained client is paired with the anchor whose update has the highest cosine similarity. The smart-matching version is the one reported as AGD-Smart.

3.3 Correction Network and Replay Buffer

The server trains a residual MLP, CorrectionNet, on pairs of quantized uploads and anchor-derived targets. The network uses one encoder layer, three residual blocks, and a zero-initialized decoder so that it starts close to the identity map and only has to learn a correction rather than a full replacement update.

Instead of training only on the current round, the server appends each new mini-batch of supervision pairs to a replay buffer capped at the most recent 10 rounds. All buffered pairs are concatenated for training. Before the correction update, buffered quantized deltas are normalized with a single scalar z-score computed across the buffer so that the network sees a more stable input scale from round to round.

3.4 Training Objective

The base training signal is a reconstruction loss that pushes corrected uploads toward their anchor-derived targets. When a server validation loader is available, the implementation can also add a distillation term. In that case, the current global model acts as the teacher, and the corrected update is also encouraged to produce similar logits on the validation split.

The teacher logits come from the current global model on the validation split. The student logits come from a model built by applying the mean corrected delta from the current round. The reported configurations use distillation temperature $\tau = 4$.

3.5 Applying Correction and Aggregation

At inference time, the server denormalizes the network output and can blend the corrected upload with the original quantized upload. The main smart-matching configuration uses an 80/20 mix of corrected and original update, which helps preserve client-specific direction under non-IID data.

The corrected deltas are then aggregated together with anchor deltas. The code supports either plain sample-count weighting or a convex combination of sample-count and cosine-similarity weights relative to the mean anchor delta. Finally, the aggregated update is applied to the global model.

3.6 Warmup

The smart-matching strategy also uses a warmup phase. For the first five rounds, the server trains the CorrectionNet and fills the replay buffer but still aggregates the raw client deltas with FedAvg. Correction is only applied after the buffer contains enough supervision signal to avoid harming early optimization.

4 Experimental Setup

Dataset and partitioning. All reported experiments use CIFAR-10. Near-IID settings use a large Dirichlet concentration ($\alpha = 100$), while the heterogeneous setting uses $\alpha = 0.5$.

Model and local training. Clients train a shared CNN for two local epochs per communication round. The server-side CorrectionNet uses hidden width 256.

Client types. Each experiment uses 10 total clients with full participation. Anchor fractions range from 20% to 50%, depending on the configuration.

Baselines. We compare against FedAvg [McMahan et al., 2017], FedShift [Seo et al., 2024], and a simplified ProWD-style baseline derived from Yoon et al. [2022]. FedShift applies a per-layer global mean shift so that the average quantized upload is aligned to the average anchor upload in the current round. The simplified ProWD baseline implemented in this repository uses a shared three-layer residual MLP trained only on the current round and targets the mean anchor delta. Because the FedShift reruns were completed on the four main reported configurations, the direct FedShift comparisons in the results focus on those shared settings.

Reported configurations. Table 1 summarizes the main settings referenced in the paper and slides.

Table 1: Reported experimental configurations.

Setting	α	Bits	Anchors	Rounds	Corr. epochs
Near-IID ProWD benchmark	100	INT8	5 / 10	50	5
Near-IID severe quant.	100	INT2	2 / 10	50	20
Near-IID moderate quant.	100	INT4	5 / 10	50	10
Non-IID heterogeneous	0.5	INT8	5 / 10	50	5

This table is mainly an interpretation guide for the later accuracy results. The near-IID INT8 row is the easiest setting and is closest to the regime where plain averaging is already strong. The INT2 row is much harsher: only two anchors supervise eight constrained clients, and each coordinate is compressed to just three effective levels. Because the rows differ in both bit-width and anchor fraction, the most important comparisons are within each row across methods, while the cross-row trend tells us when correction becomes necessary.

Evaluation. The primary metric is top-1 accuracy on the global CIFAR-10 test set at selected communication rounds. We focus on round 10 to measure early usefulness and round 50 to measure whether gains persist.

5 Results

5.1 Main Result: INT2 Is the Regime Where Correction Matters

Figure 2 shows the strongest result in the project: near-IID CIFAR-10 with INT2 quantization and only 20% anchors. This is the regime where plain averaging struggles most because each uploaded coordinate has only three effective levels.

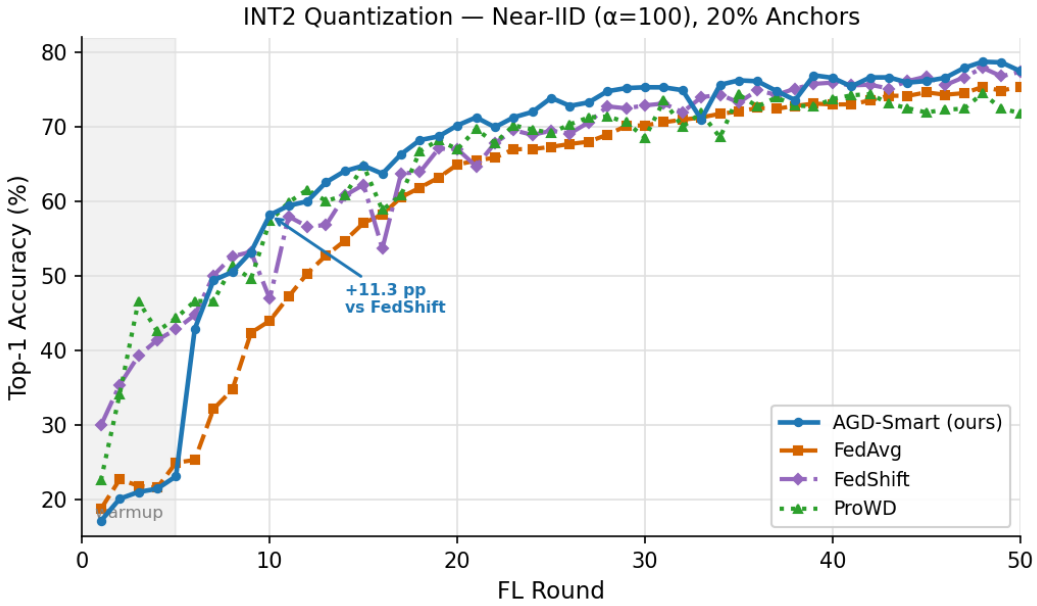


Figure 2: INT2 near-IID benchmark over 50 rounds. AGD-Smart gains sharply after warmup and stays ahead of FedAvg throughout the full run.

Table 2: INT2 near-IID checkpoints (%).

Method	R10	R20	R30	R50
FedAvg	44.0	64.9	70.2	75.4
FedShift	46.9	67.0	72.9	77.4
ProWD	57.3	67.0	68.5	71.8
AGD-Smart	**58.2**	**70.2**	**75.3**	**77.5**

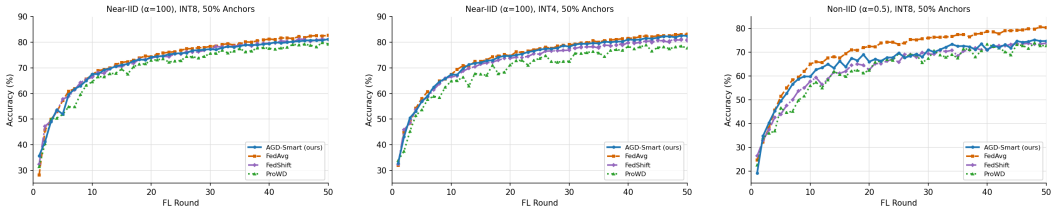
Table 2 shows that AGD-Smart is the best method at every reported checkpoint, but the comparison to FedShift changes the interpretation of the result. FedShift already fixes much of the end-of-training bias: by round 50 it reaches 77.4, essentially tied with AGD-Smart at 77.5 and clearly above FedAvg. The remaining advantage of AGD-Smart is therefore front-loaded. It leads FedShift by +11.3 points at round 10 and by +3.2 at round 20, which is exactly the interval where a communication-limited system benefits most from a stronger correction policy. Relative to ProWD, the gap still widens over time, reaching +5.7 at round 50. That separates the three correction strategies clearly: ProWD forgets too much from round to round, FedShift eventually recovers but reacts slowly, and AGD-Smart combines fast activation with strong final accuracy.

5.2 Noise Scaling

Table 3 shows that the method's value is highly dependent on quantization severity. Once the bit-width rises to INT3 or above, FedAvg is usually good enough. The correction pipeline becomes most valuable when the quantized representation collapses too much directional information. FedShift was rerun only on the four shared headline configurations, so this table keeps the broader bit-width threshold comparison against FedAvg and ProWD.

Table 3: Round-20 accuracy under different near-IID bit-widths with 20% anchors.

Bits	Levels	FedAvg	ProWD	AGD-Smart	vs. FedAvg
INT8	255	72.0	66.4	68.2	−3.9
INT6	63	73.0	68.3	70.7	−2.2
INT4	15	74.5	65.4	72.8	−1.7
INT3	7	74.4	65.8	71.7	−2.8
INT2	3	65.4	59.8	**70.7**	**+5.3**



(a) Near-IID, INT8 (b) Near-IID, INT4 (c) Non-IID, INT8

Figure 3: Secondary regimes. The method stays consistently above ProWD, nearly matches FedAvg under near-IID INT4, and narrows the gap to FedAvg under non-IID INT8 without fully closing it.

Table 3 makes the crossover point very clear. Compared with ProWD, AGD-Smart wins at every bit-width, from +1.8 points at INT8 up to +10.9 points at INT2, so the improved correction design is consistently better than the simpler learned baseline. Compared with FedAvg, however, the sign of the gain flips with quantization severity: AGD-Smart is still 1.7–3.9 points behind from INT8 through INT3, then jumps to a +5.3 advantage at INT2. That pattern supports the main claim of the paper. Learned dequantization is not universally necessary, but it becomes valuable once the quantizer is coarse enough that plain averaging cannot preserve update direction. The shared FedShift reruns follow the same story: it is competitive in the easier near-IID settings and helpful at INT2, but the learned model becomes more valuable once early convergence or heterogeneous matching matters.

5.3 Behavior in Milder Regimes

Figure 3 summarizes the remaining reported settings. The pattern is consistent: AGD-Smart reliably beats ProWD, but it only surpasses FedAvg once quantization is severe enough that averaging alone can no longer recover the signal.

Table 4 sharpens the baseline story. FedShift is the stronger lightweight comparator: it beats ProWD in three of the four shared settings and nearly matches AGD-Smart at the easier near-IID endpoints. That makes the INT4 and non-IID rows more informative than the near-IID INT8 tie. At INT4, AGD-Smart keeps a +1.8 final advantage over FedShift while finishing within 0.7 of FedAvg. Under non-IID INT8, the final gap is +1.0 and the midpoint gap is +3.9 at round 25, which suggests that smart client-to-anchor matching matters more than a single global shift once anchor and constrained populations no longer share the same mean direction. INT2 is different again: the final accuracy is nearly tied, but the route to that endpoint is not.

Figure 4 makes that contrast explicit. FedShift is already a very strong no-learning baseline when the client population is near-IID. At near-IID INT8 and INT2, the final gap is only +0.1. But those near ties hide where AGD-Smart helps most. In the severe INT2 regime it opens an 11.3-point lead by round 10, which matters if communication rounds are scarce. At INT4, it keeps a meaningful +1.8 end-of-training advantage, suggesting that the replayed learned corrector is still buying something after simple mean alignment has done its part. Under non-IID data, FedShift loses more clearly because the anchor mean is no longer a reliable global target.

Table 4: Summary across shared configurations, including FedShift.

Setting	FedShift @ R50	AGD-Smart @ R50	vs. FedShift	vs. ProWD	vs. FedAvg
Near-IID, INT2, 20% anchors	77.4	77.5	+0.1	+5.7	+2.1
Near-IID, INT8, 50% anchors	81.1	81.2	+0.1	+1.9	-1.5
Near-IID, INT4, 50% anchors	80.7	82.5	+1.8	+4.8	-0.7
Non-IID, INT8, 50% anchors	73.8	74.8	+1.0	+2.0	-5.6

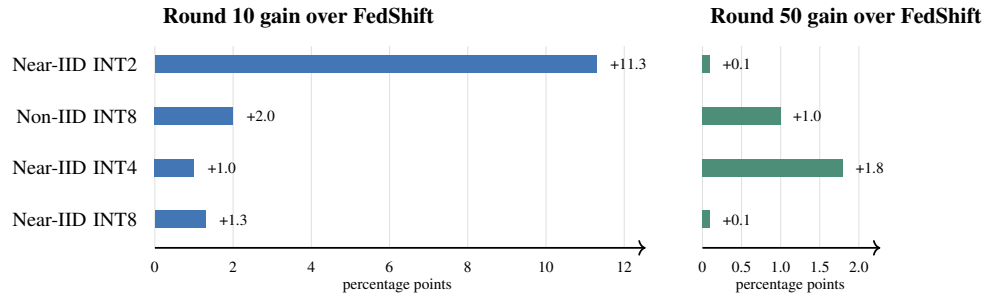


Figure 4: AGD-Smart's gain over FedShift at round 10 and round 50 on the four shared configurations. The largest early-round advantage appears under INT2, while the largest persistent end-of-training gain appears at INT4.

6 Conclusion

Using the provided course template, this paper reframes the project as a server-side correction pipeline for bit-width-heterogeneous federated learning. The central result is not that correction universally beats FedAvg. It does not. Instead, the evidence points to a narrower and more useful claim: anchor-guided correction is most valuable when quantization is so aggressive that averaging cannot reliably reconstruct the true direction of the update.

In that regime, especially INT2, AGD-Smart provides substantial early-round and end-of-training gains while remaining consistently stronger than the simplified ProWD baseline in every reported configuration. FedShift sharpens that story: a simple mean-shift baseline already recovers much of the easy near-IID performance, but AGD-Smart still wins on early INT2 convergence, on final INT4 accuracy, and under non-IID data where a single global shift is too coarse. The reusable method diagram and the revised slides emphasize the practical mechanics that drive that result: anchor supervision, replay buffering, smart matching, warmup, and blended correction.

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